Introduction

Population genetics quantifies genetic variation within and between populations. Allele frequencies, Hardy-Weinberg equilibrium (HWE), and F-statistics form the foundational toolkit for diagnosing genotyping errors, detecting population structure, and assessing selection or drift.

Prerequisites

Mendelian inheritance; genotype coding (0/1/2 for minor-allele dosage).

Theory

Allele frequency. For a biallelic SNP with minor-allele counts n_{AA}, n_{Aa}, n_{aa}, $p = (2n_{AA} + n_{Aa})/(2n)$.

Hardy-Weinberg equilibrium (HWE). Under random mating and no selection, expected genotype frequencies are $p^2, 2pq, q^2$. Exact HWE tests (`HWExact`) are preferred over the chi-square for rare variants.

F-statistics. $F_{ST} = (H_T - H_S)/H_T$ partitions heterozygosity between and within populations, quantifying structure (Wright, 1951). Typical human F_{ST} between continents ≈ 0.1.

Assumptions

Random mating, no selection, no mutation, no migration, infinite population size; genotyping is unbiased.

R Implementation

```r
library(HardyWeinberg)

set.seed(2026)
# Simulate HWE: p = 0.3, n = 200 diploid individuals
p <- 0.3
n_AA <- rbinom(1, 200, p^2)
n_aa <- rbinom(1, 200, (1 - p)^2)
n_Aa <- 200 - n_AA - n_aa

gt <- c(AA = n_AA, AB = n_Aa, BB = n_aa)
HWExact(gt, verbose = TRUE)

# Simulate HWE violation (excess heterozygotes)
gt_violate <- c(AA = 20, AB = 160, BB = 20)
HWExact(gt_violate, verbose = TRUE)

# F_ST between two populations (manual calculation)
p1 <- 0.2; p2 <- 0.5
p_bar <- mean(c(p1, p2))
H_T <- 2 * p_bar * (1 - p_bar)
H_S <- mean(c(2 * p1 * (1 - p1), 2 * p2 * (1 - p2)))
Fst <- (H_T - H_S) / H_T
Fst
```

Output & Results

HWE exact test p-value (large for the simulated HWE case; ~0 for the violated case). $F_{ST} \approx 0.1$ for the two-population toy example.

Interpretation

"HWE exact tests excluded 1,240 variants ($p < 1e\text{-}6$) from association analysis as likely genotyping artefacts. Between-cohort $F_{ST} = 0.012$ indicates minimal population stratification after ancestry matching."

Practical Tips

- Always HWE-filter variants in GWAS pre-processing; strong HWE violations usually indicate genotyping errors.
- Test HWE in controls only (not cases) because true disease associations can cause HWE departures in cases.
- For low-MAF variants (<5%), asymptotic chi-square is unreliable; use the exact test.
- `adegenet` and `hierfstat` provide comprehensive F-statistics and population-structure utilities.
- F_{ST} between samples can be inflated by small sample size; apply Weir-Cockerham estimator and bootstrap CIs.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- RNA-seq with DESeq2 / edgeR
- Enrichment analysis (GSEA / ORA)
- scRNA-seq with Seurat